

PROJECT PLAN

Improved Library Discovery and Management System for WG Solheim Library

Group ACF

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Client: WG Solheim Library

Project Website: [TBA]

CUSTOMER STATEMENT OF REQUIREMENTS

Project Background

Libraries provide users with access to a large amount of information, including books, journals, theses, electronic resources, and other materials. UP Libraries' TUKLAS serves as a centralized discovery service that brings together information from various local library databases and electronic resources into a single search platform.

During consultations with WG Solheim Library, staff highlighted significant operational challenges with relying directly on TUKLAS for primary data entry. Limited time and tight deadlines often force staff to rush cataloging and data entry tasks. Consequently, library staff strongly prefer having their own standalone, private database and dedicated website.

A dedicated system allows staff to create, edit, and maintain records at their own pace without operational pressure, while retaining the capability to export and import updated catalog data to TUKLAS whenever necessary.

Furthermore, as a specialized archaeology library, WG Solheim Library has unique resource management needs that general discovery platforms do not fully address. The system must accommodate high-resolution artifact images and rich, detailed archaeological descriptions alongside traditional book and periodical collections.

Additionally, the library requires a customized borrowing database system tailored specifically to their day-to-day circulation workflows and operational requirements.

Current Practice

The current practice relevant to this project involves the use of TUKLAS as a reference point for discovering library resources.

TUKLAS provides a centralized search platform for library resources. Rather than requiring users to search through multiple sources individually, it integrates information from local databases and electronic resources into a single platform. UP Libraries describes TUKLAS as a discovery tool that can search across different collections and provide bibliographic information to users.

This approach is useful because it gives users a single starting point when looking for library resources.

However, the initial consultation with WG Solheim Library identified the user interface and usability of the current approach as an issue. The design can make it difficult for users to locate information efficiently.

The problem can therefore be viewed from two perspectives:

1. Information availability — users need access to the library's available resources and their relevant information.
2. Information accessibility — users need to be able to understand and navigate the system easily enough to locate those resources.

The proposed project focuses primarily on the second concern while retaining the useful discovery functionality expected from a library system.

Proposed Solution

The group proposes the development of an improved library discovery and management system for WG Solheim Library.

The system will provide functionality similar to the discovery capabilities of TUKLAS while introducing a redesigned user interface and improved navigation. The goal is to make common library-related tasks easier to understand and complete.

The proposed system will initially focus on the following areas:

Resource Search. Users should be able to search the library's available resources through a centralized search interface.

Resource Information. Search results should present relevant information about each resource in an understandable format. Depending on the final requirements, this may include information such as title, author, subject, publication information, location, and availability.

Search Refinement. Users should be able to narrow their results when a search produces a large number of results.

Resource Discovery. The interface should allow users to browse and discover resources beyond a single exact search.

Improved Navigation. The system should provide clear navigation paths so that users can understand where they are within the system and how to return to previous results or sections.

Library Management & Private Database. Library staff will have an independent private database and website to manage records without rush, allowing data entry at their own pace and batch exporting/importing to TUKLAS when ready. The system will also feature a customized borrowing database system.

Archaeological Artifact Support. To support the specialized needs of an archaeology library, the system will accommodate artifact images and rich detailed descriptions alongside standard book resources.

The exact scope of administrative functionality will be determined through further consultation with WG Solheim Library.

Project Goals

The primary goal of the project is to develop a library system that allows users to find and access library information more efficiently than through the current interface.

The project has the following goals:

1. Improve the overall usability of the library system.
2. Create a clearer and more intuitive user interface.
3. Make library resources and archaeological artifacts (with images and detailed descriptions) easier to search and discover.
4. Organize resource information in a way that is easier to understand.
5. Provide clear navigation between major sections of the system.

6. Provide appropriate management functionality for library staff if required by the final project scope.
7. Provide a dedicated private database allowing library staff to update catalog data at their own pace and import to TUKLAS as needed, along with a customized borrowing management system.
8. Develop a maintainable system that can be managed and modified more easily.
9. Validate the proposed system through testing and feedback from the client.

Target Users

The exact user roles have not yet been finalized with WG Solheim Library.

For preliminary planning purposes, the system is expected to support two broad groups:

Library Users. These may include students, faculty, researchers, and other individuals who need to search for library resources.

Library Staff. Library staff may require administrative functionality for managing library resources and system information.

These user categories are preliminary and will be validated during subsequent client consultations.

Preliminary Requirements

Because the project is still in its early stages, the following requirements represent the team's **preliminary requirements**. Requirements that depend on further client clarification are subject to change.

Functional Requirements

Resource Search. The system shall allow users to search for library resources using relevant search terms.

Search Results. The system shall display search results in an organized and understandable format.

Resource Details. The system shall allow users to view detailed information about a selected resource.

Resource Availability. The system shall display the availability or status of a resource when such information is available.

Resource Location. The system shall provide the location of physical resources when applicable.

Search Refinement. The system shall allow users to refine or narrow search results.

Resource Browsing. The system shall provide users with ways to browse available resources.

Navigation. The system shall provide clear navigation between the major pages and functions of the system.

External Resource Access. When applicable, the system shall provide links to external websites or databases where electronic resources can be accessed.

Artifact Cataloging. The system shall support high-resolution artifact images and detailed archaeological descriptions alongside traditional book records.

Customized Borrowing System. The system shall include a customized borrowing database system to track circulation of library items and artifacts.

TUKLAS Data Import/Export. The system shall allow staff to work on their local private database and export/import updated records to TUKLAS when ready.

Resource Management. The system shall provide authorized library personnel with functionality for managing resource information, subject to confirmation from the client.

User Management. If required by the client, the system shall provide authorized personnel with functionality for managing user information.

Authentication. If administrative functionality requires authentication, the system shall restrict administrative features to authorized users.

Data Management. The system shall store and retrieve library information in an organized manner.

Error Handling. The system shall provide understandable feedback when a requested resource cannot be found or an operation cannot be completed.

Responsive Interface. The system should provide a usable interface across commonly used screen sizes.

Non-Functional Requirements

Usability. The system should be understandable to users with varying levels of familiarity with online library systems.

Performance. Searches and common user actions should provide results within a reasonable amount of time.

Maintainability. The system should be organized so that future developers or library personnel can maintain and update it.

Reliability. The system should consistently provide correct resource information based on the data available to it.

Security. Administrative functions and other sensitive operations should be protected against unauthorized access.

Accessibility. The interface should use readable text, clear controls, and understandable navigation.

Compatibility. The system should operate using commonly available modern web browsers if a web-based implementation is selected.

Preliminary UI Design

The preliminary interface will prioritize simplicity and ease of navigation rather than reproducing the existing TUKLAS interface exactly.

The following screens are proposed for the initial design.

Home/Search Page. The home page will provide users with a prominent search bar as the primary method of accessing library resources. Possible elements include:

- Library branding
- Search bar
- Search button
- Search category/type selector
- Navigation menu
- Links to browsing or featured resources

Search Results Page. After performing a search, the user will be shown a list of matching resources. Possible elements include:

- Search query
- Number of results
- Search filters
- Sorting options

- Resource cards/list items
- Availability indicator
- Pagination or continuous loading

Resource Details Page. Selecting a resource will open a page containing more detailed information. Possible information includes:

- Title
- Author
- Publication information
- Subject
- Resource type
- Availability
- Location
- Call number
- Links to electronic resources where applicable
- Artifact images and archaeological descriptions

Management Dashboard. If administrative functionality is confirmed, library staff will have access to a dashboard for managing system information. Possible functions include:

- Resource management
- User management
- Search/browse management
- System information
- Reports
- Customized borrowing database system
- TUKLAS import/export module

These features remain subject to client confirmation.

Preliminary Navigation Flow

The basic user navigation is expected to follow:

Home → Enter Search Query → Search Results → Apply Filters / Sort Results → Select Resource → Resource Details → View Location / Availability / External Resource →

For administrative users, a preliminary path is:

Login → Management Dashboard → Select Management Function → View / Add / Edit / Remove Information → Save Changes

The exact navigation structure will be revised after additional client meetings and UI prototyping.

References and Existing Solutions

The primary existing solution used as a reference for this project is TUKLAS: UP Libraries' Discovery Service. TUKLAS combines library database sources and electronic resources into an all-in-one search platform.

UP Libraries also documents TUKLAS as a discovery tool based on VuFind and customized for the information-seeking needs of the UP community. Its stated functions include searching, discovering, and accessing information resources.

The project will also use *Software Engineering: A Practitioner's Approach* by Roger S. Pressman and Bruce R. Maxim as a software-engineering planning reference. The book covers project planning, project metrics, estimation, scheduling, and risk management as part of its project-management material.

PLAN OF WORK

Estimate Cost and Resources

The project is planned as a year-long academic software engineering project. Since the final technical architecture and deployment requirements have not yet been determined, the following estimates are preliminary.

The project primarily requires human resources, development resources, computing resources, and communication resources.

Human Resources. The project team consists of three student developers:

Member	Preliminary Responsibility
Renson Nicolai Villareal	Project coordination, development, testing
John Ray Donovan Añora	Development, UI/UX, testing

Ronaldvince Yrlwin Aquino	Development, database/system design, testing
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The exact division of responsibilities will be finalized after the system architecture and modules have been established.

Software Resources. The project may require:

- Code editor/IDE
- Git
- GitHub
- Web browser
- Database management software
- UI/UX design software
- Testing tools
- Development frameworks

The exact programming language, framework, database system, and deployment platform are **to be determined**.

Hardware Resources. Each developer is expected to require:

- Personal computer capable of running the development environment
- Minimum 8 GB RAM recommended
- Sufficient local storage for source code, dependencies, and project data
- Stable internet connection

These specifications are preliminary and may be revised according to the selected technology stack.

Communication Resources. The project requires:

- Email communication with the client
- Online or face-to-face client meetings
- Group communication platform
- GitHub for source-code collaboration

Estimation Techniques

Two estimation approaches will initially be used:

1. Problem-Based Estimation

2. Process-Based Estimation

Using multiple estimates provides an opportunity to compare the expected effort from different perspectives before committing to a final project schedule. Pressman's project-estimation material discusses decomposition techniques such as problem-based and process-based estimation and recommends reconciling estimates rather than relying on a single estimate.

Problem-Based Estimation. The project will first be decomposed into major functional areas.

Functional Area	Preliminary Effort
Requirements gathering	60 hours
UI/UX design	100 hours
Search and discovery	140 hours
Resource information	100 hours
Database/data management	120 hours
User/admin functionality	100 hours
Integration	80 hours
Testing	120 hours
Documentation	80 hours
Deployment	40 hours
Estimated Total	940 hours

These numbers are initial planning estimates, not commitments. They will be revised after requirements and architecture have been established. With three members, this corresponds to approximately 313 person-hours per member over the full project period.

Process-Based Estimation. The second estimate will be based on development activities rather than individual features.

Development Activity	Estimated Effort
Project planning	50 hours
Requirements engineering	100 hours
UI/UX design	100 hours
System/database design	100 hours
Implementation	300 hours
Integration	70 hours
Testing and debugging	120 hours
Deployment	40 hours
Documentation	80 hours
Estimated Total	960 hours

The difference between the two preliminary estimates is 20 hours or approximately 2.1% of the problem-based estimate. The two estimates are therefore reasonably close for an initial project plan. The team will reconcile these estimates as the requirements become clearer.

Project Cost. Because this is an academic project, the team does not currently anticipate major direct development costs.

Resource	Estimated Cost
Development hardware	Existing team resources
Development software	₱0 / existing or open-source tools
GitHub	₱0 for planned use
Communication	Existing resources

Hosting	TBD
Database/server	TBD
Other services	TBD

The current expected direct project cost is therefore minimal, although hosting, server infrastructure, domain registration, or other services may introduce costs later.

System Resources

The proposed system will require the following resources.

Display. The system should support common desktop and laptop displays and should be designed responsively for smaller screens.

Storage. Storage requirements will depend on the final database and whether the system stores only bibliographic information or additional digital resources.

Network. The system will require an internet or local network connection depending on the final deployment architecture.

Server. If the system is deployed as a web application, it will require a server or hosting environment capable of running the selected application framework and database.

Database. A database will be required to store information such as resources, users, availability, and other data determined during requirements analysis.

The exact database technology is **to be determined**.

Project Schedule

The project is expected to span approximately one academic year. Because the project has only recently started, the following schedule is preliminary.

Phase	Target Period	Major Deliverable
Project initiation	August–September 2026	Initial project plan
Requirements gathering	September–October 2026	Requirements specification
Client validation	October 2026	Confirmed requirements

UI/UX design	October–November 2026	Wireframes and prototypes
System design	October–November 2026	Architecture/database design
Initial implementation	November–December 2026	Prototype
Feature implementation	November–December 2026	Functional system
Integration	February 2027	Integrated system
Testing	February - April 2027	Analysis and Refinement
Client evaluation	April 2027	Client feedback
Refinement	April - May 2027	Revised system
Final documentation	May 2027	Final documentation

The schedule will be revised once the course's official project deadlines and the client's availability are known.

Major Milestones

Project Plan Completed. Initial project plan and scope documented.

Requirements Baseline. Major functional and non-functional requirements validated with the client.

UI Prototype Completed. Initial interface design completed and presented for client feedback.

System Design Completed. Database and software architecture established.

Functional Prototype. Core search and resource discovery functionality implemented.

Feature-Complete System. Major agreed-upon features implemented.

Integrated System. Components integrated into a single working system.

System Testing Completed. Major defects identified and addressed.

Client Acceptance Testing. Client evaluates the system against agreed requirements.

Final System. Final refinements and documentation completed.

Project Metrics

Project metrics will be used to monitor progress and identify potential delays. This is consistent with the project-management emphasis in *Software Engineering: A Practitioner's Approach*, where project metrics are used to monitor project status and support corrective decisions.

The following metrics will be monitored.

Schedule Progress

Schedule Completion Percentage

Schedule Completion = Completed Tasks/Planned Tasks

This will show how much of the planned work has been completed.

Effort

The team will record the approximate number of hours spent on major tasks.

This will allow actual effort to be compared with the original estimate.

Requirement Completion

Requirement Completion = Completed Requirements/Total Approved

This will indicate whether implementation is keeping up with the requirements baseline.

Defect Count. The team will track:

- Newly discovered defects
- Resolved defects
- Remaining defects
- Severity of defects

Task Completion. Each task will be tracked as:

- Not started
- In progress
- Completed
- Blocked

Client Feedback. The team will record feedback from client meetings and classify it according to:

- New requirement
- Usability issue

- Functional issue
- Change request
- Clarification

These metrics will be reviewed during regular team meetings.

Effort Breakdown

The team currently consists of three members. Since specific roles have not yet been assigned, responsibilities will initially be divided according to major project areas.

Project Area	Renson Nicolai Villareal	John Ray Donovan Añora	Ronaldvince Yrlwin Aquino
Requirements	✓	✓	✓
Client coordination	✓	✓	✓
UI/UX design	✓	✓	
Front-end development	✓	✓	
Back-end development	✓		✓
Database		✓	✓
Search functionality	✓		✓
Testing	✓	✓	✓
Integration	✓	✓	✓
Documentation	✓	✓	✓
Project coordination	✓		

This allocation is preliminary and will be updated once the team determines its actual roles and system architecture.

Integration Coordinator. The team will designate one member as the primary integration coordinator after the system architecture is finalized.

Integrated System Tester. Testing responsibilities will be shared among the three members, with one member eventually designated to coordinate integrated-system testing.

Development Workflow

The team will follow an iterative development approach.

A preliminary cycle will be:

Requirements → Design → Implementation → Testing → Client Feedback → Refinement

This allows the team to identify usability issues early rather than waiting until the end of the project.

Client feedback will be incorporated into subsequent iterations when the requested changes are consistent with the project's agreed scope.

PROJECT RISKS

The project will use a risk table to identify, assess, and plan responses to potential problems. Risk management is particularly important because the project has an extended development period and its requirements are not yet fully defined. Pressman's risk-management material emphasizes identifying risks, assessing their probability and impact, and planning mitigation and monitoring strategies.

Risk Table

Risk	Probability	Impact	Risk Level	Mitigation
Requirements may change during development	High	High	High	Conduct regular client meetings and maintain a requirements baseline

Client availability may be limited	Medium	High	High	Schedule meetings in advance and maintain communication through email
Team members may have conflicting academic schedules	High	Medium	High	Divide tasks clearly and maintain a shared schedule
Project scope may become too large	High	High	High	Prioritize core features and establish a clear scope
Team may underestimate development effort	Medium	High	High	Compare multiple estimation methods and revise estimates periodically
Technical stack may be selected too late	Medium	Medium	Medium	Establish technical requirements after initial requirements analysis
Integration problems may occur	Medium	Medium	Medium	Use Git and integrate components regularly
Bugs may accumulate near the deadline	Medium	High	High	Perform testing throughout development instead of only at the end
UI improvements may not satisfy actual users	Medium	High	High	Conduct usability testing and obtain client/user feedback
Library data may be incomplete or unavailable	Medium	High	High	Determine available data sources early and establish a test dataset

Security problems may occur in administrative features	Medium	High	High	Apply authentication and authorization practices and conduct security testing
Team member becomes unavailable	Low	High	Medium	Document work and avoid assigning critical knowledge to only one person
Deployment environment may not support the selected system	Medium	Medium	Medium	Identify deployment requirements before implementation is completed
Schedule may slip because of delayed milestones	Medium	High	High	Monitor schedule metrics and address blocked tasks immediately
The final system may become difficult to maintain	Medium	Medium	Medium	Use organized code, documentation, version control, and consistent development practices

High-Priority Risks

Requirements Uncertainty. The project currently has only one client consultation, meaning that several requirements have not yet been confirmed. This is one of the project's highest risks because implementing incorrect assumptions could result in significant rework. The team will conduct additional client meetings, document requirements, and obtain confirmation before treating major features as finalized.

Scope Expansion. Because TUKLAS provides many discovery capabilities, attempting to reproduce every function could make the project unnecessarily large. The team will establish a minimum viable feature set and prioritize features based on client and user needs.

Client Availability. Limited opportunities to communicate with the client may delay requirements validation and feedback. The team will schedule meetings ahead of time and use email for questions that do not require a full meeting.

Schedule Delays. The project will run alongside the team's other academic responsibilities. The team will track task completion, effort, and milestone progress. Delayed tasks will be identified during regular project reviews.

Usability Failure. The project's primary motivation is to improve usability. A technically functional system would still fail to meet the project's primary objective if users find it difficult to use. The team will conduct usability evaluation throughout development and use client/user feedback to refine the interface.

Risk Monitoring

The team will review project risks throughout development rather than only during the initial planning stage.

For each major risk, the team will monitor:

- Whether the risk has become more or less likely
- Whether its potential impact has changed
- Whether mitigation actions are working
- Whether a contingency action is necessary

New risks identified during development will be added to the risk table.

CONCLUSION

The proposed project aims to develop an improved library discovery and management system for WG Solheim Library. The project is motivated by the difficulty users may experience when finding and accessing information through the current TUKLAS interface.

Rather than simply reproducing TUKLAS, the proposed system will use its general discovery functionality as a reference while focusing on improved usability, navigation, organization, and manageability.

Because the project is still in its early stages, several aspects—including the final feature set, user roles, technical architecture, database, and deployment environment—remain subject to client consultation and requirements analysis.

The project plan therefore serves as an initial baseline. As the team gathers additional information from WG Solheim Library, the requirements, estimates, schedule, responsibilities, and risks will be refined accordingly.

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